

OUTLOOK INTEGRATED SMART AI PROJECT MANAGEMENT DASHBOARD



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A project proposal submitted for
Final Year Project

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DECLARATION

We declare that this project proposal is based on our original work, except for citations and quotations that have been properly acknowledged. We also declare that it has not been submitted previously or at the same time for any other degree or award at Bahria University or another institution.

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APPROVAL FOR SUBMISSION

I certify that the project proposal entitled “OUTLOOK INTEGRATED SMART AI PROJECT MANAGEMENT DASHBOARD” was prepared by Zain Ahmed Siddiqui, Syed Faaiz Noman, and Zoha Asim. The proposal has met the required standard for submission in partial fulfilment of the requirements for the award of Bachelor of Science in Artificial Intelligence at Bahria University.

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OUTLOOK INTEGRATED SMART AI PROJECT MANAGEMENT DASHBOARD

ABSTRACT

Effective project management depends on the timely identification, interpretation, and monitoring of meetings, deadlines, document requests, responsibilities, and follow-up actions. In many organizations, such information remains scattered across lengthy Microsoft Outlook email threads and is manually transferred into Microsoft Excel, resulting in fragmented records, duplicated effort, delayed responses, and missed commitments.

This study proposes an AI-enabled project management software application designed to operate within an existing Microsoft Outlook 2016 and Microsoft Excel environment. The proposed system adopts a hybrid architecture that combines a lightweight NLP pipeline for email classification and information extraction with a rule-based verification layer for validating dates, senders, recipients, attachments, thread relationships, and organizational patterns. It will classify project-related emails and extract structured information, including meeting subjects, dates, times, participants, agendas, responsible individuals or departments, deadlines, action items, and requested documents.

The system will also analyze related messages within the same thread to detect revised deadlines, identify changes in responsibility, update existing records, reduce duplicate entries, and prioritize pending actions. Extracted information will be stored in a structured Excel archive and presented through a centralized desktop dashboard that supports deadline tracking, document monitoring, departmental accountability, and project status summaries. Automated alerts will notify users about upcoming meetings, overdue tasks, missing minutes, reports, presentations, approvals, and pending inputs.

The expected outcome is an AI-enabled project management system that reduces manual administrative work, improves record accuracy, strengthens follow-up practices, and supports timely task completion without disrupting established organizational workflows.

1.1 INTRODUCTION & BACKGROUND

Project communication is often spread across meeting invitations, replies, forwarded messages, deadline changes, and requests from different departments. Outlook stores these messages, but it does not convert them into one clear project record. A manager still has to open several threads to understand what is planned, who is responsible, and what has not been delivered.

The proposed solution extends traditional reminder systems by incorporating email classification and information extraction. The AI processing module must classify semi-structured and unstructured email content, identify the type and purpose of each message, extract structured project information, and determine what action should follow. It must also preserve a formal record in Microsoft Excel and present the information through a centralized dashboard.

The system uses contextual analysis instead of relying only on fixed keywords. It will connect related messages in the same email thread, recognize when a deadline has changed, link an action item with its responsible person, identify missing post-meeting documents, and place urgent follow-ups above routine updates. The manager will also be able to use the dashboard search to retrieve a status view of meetings, deadlines, and pending departmental inputs from the stored project archive.

A hybrid design is suitable for this work. A lightweight NLP pipeline will handle email classification, entity extraction, and action item detection, while distinct rules will verify exact fields such as dates, email metadata, attachment names, and project-specific terms. This combination keeps the system practical for office computers and gives the user more control over the extracted record [1], [2].

1.2 AIM OF THE PROJECT

The aim of this project is to develop a stand-alone AI-enabled project management software application that integrates with Microsoft Outlook 2016, classifies project emails, extracts structured information, creates an organized archive of meetings and deadlines in Microsoft Excel,

and supports timely follow-up through a centralized desktop dashboard, automated reminders, and local multi-user access.

1.3 PROBLEM STATEMENT

Project managers receive important instructions through email, but the information is scattered across separate messages and long conversation threads. Meeting dates, revised timelines, requested reports, responsible departments, and pending inputs are often copied into spreadsheets or personal notes. This manual process is slow, depends on regular checking, and makes it difficult for senior managers to see the current project position at a glance.

Existing tools usually solve only one part of the problem. Outlook manages communication, Excel stores records, and calendar reminders notify users about dates. They do not work together as a single system that classifies email content, extracts project information, updates records when conversations change, and identifies missing documents after a meeting or deadline.

The proposed system addresses this gap by combining Outlook email monitoring, NLP-based email classification and information extraction, rule-based verification, Excel archival, structured follow-up logic, and a centralized dashboard in one local desktop software application.

1.4 OBJECTIVES

The project has four objectives:

1. Develop an AI-based email processing module that connects with Outlook 2016 and classifies project-related messages, including their type, origin, responsible person, timeline, meeting details, and action items.
2. Create a centralized project record by logging extracted meetings, deadlines, requested documents, pending inputs, and follow-up status in Microsoft Excel and presenting them on a desktop dashboard.

3. Provide structured monitoring through thread-aware updates, deadline and priority assignment, missing-document tracking, automated alerts, and dashboard search and filtering over the stored archive.
4. Provide custom settings and controlled single-user and multi-user operation within a local network without forcing managers to change their existing Outlook and Excel working style.

1.5 PROJECT SCOPE

The project covers a Windows desktop application that communicates with Microsoft Outlook 2016 and Microsoft Excel. It will monitor selected Outlook folders and process English-language project emails related to meetings, deadlines, reports, presentations, approvals, departmental inputs, action items, and follow-up requests.

The AI processing module will analyze the subject, message body, sender, recipients, date, thread history, and attachment names. It will classify the email type, identify the origin, actioner, timeline, meeting details, requested documents, and pending actions through named entity recognition and information extraction. It will also compare related messages so that revised dates or changed responsibilities update the existing project record instead of creating a separate disconnected entry.

The centralized dashboard will show upcoming meetings, approaching deadlines, overdue actions, missing minutes of meeting, pending reports, and departmental inputs. A dashboard search and filtering area will allow users to retrieve status information from the stored archive, including queries such as “What is due this week?” and “Which departments have not submitted their inputs?”

The first release will support a single user. Multi-user access will be added within a local network with user roles and controlled access to the shared archive. The system will record attachment names but will not analyze the full content of every attached file in the first version. It

will support follow-up prompts and reminders, but final outgoing emails will remain under the user's control.

1.6 OUTCOME

The project is expected to produce four outcomes:

1. A working AI-enabled project management software application that automatically detects Outlook 2016 project emails and extracts the required structured information.
2. An Excel-based formal archive and centralized desktop dashboard for meetings, deadlines, action items, requested documents, pending inputs, and follow-up history.
3. An automated alert and follow-up system that handles revised deadlines, highlights urgent items, tracks missing post-action documents, and provides dashboard-based project status retrieval.
4. A configurable software package that supports individual use and controlled multi-user access on a local network.

Direct beneficiaries are project managers, senior managers, and team leads who need a clear record of meetings and deadlines. Departments and team members are indirect beneficiaries because the system will send earlier reminders and reduce repeated manual follow-up for minutes, reports, presentations, and pending inputs.

BACKGROUND / LITERATURE REVIEW

This chapter reviews recent work on enterprise email processing, event extraction, temporal understanding, action item extraction, information extraction, lightweight NLP techniques, and AI-supported project management. The selected studies support the proposed system's email classification, named entity recognition, information extraction, action item extraction, deadline detection, and project monitoring capabilities.

2.1 BACKGROUND

Recent research shows that an effective email processing system needs more than one NLP function. It must classify the message's purpose, extract people and events, interpret dates in context, combine information from long conversations, and present the result in a form that supports decisions. Table 2.1.1 summarizes the studies used to shape the proposed design.

Table 2.1.1: Literature Review

Year	Technology / Study	Application	Dataset	Evaluation Metrics	Relevance to Proposed Project	Ref.
2023	MailEx: email event and argument extraction	Extracts events and related roles from conversational email threads	1.5K email threads, about 4K emails and about 8K event instances	Event trigger and argument extraction F1	Supports extraction of meetings, actions, responsible people and thread-level context from email	[1]
2025	Advanced Messaging Platform (AMP)	Enterprise email parsing, intent recognition, entity extraction, actions and human validation	92,764 private email conversations; three operations datasets; Banking77; 200 annotated pipeline emails	Parser accuracy, precision, recall, F1 and draft-use rate	Closest example of a complete enterprise AI email pipeline and hybrid human review	[2]

Year	Technology / Study	Application	Dataset	Evaluation Metrics	Relevance to Proposed Project	Ref.
2024	Discourse-aware temporal normalization	Converts phrases such as “next Friday” into standard dates using surrounding context	TempEval3, i2b2, AncientTimes and TempCourt across six domains and seven languages	Temporal normalization accuracy	Supports correct interpretation of relative deadlines and meeting dates across email threads	[3]
2023	MeetingBank benchmark	Creates meeting summaries aligned with agendas and professional minutes	1,366 meetings, 3,579 hours of video and 6,892 segment-level instances	ROUGE scores and expert human assessment	Supports concise meeting records and post-meeting document tracking	[4]
2022	Focus on the Action: Learning to Highlight and Summarize Jointly for Email To-Do Items Summarization	Automatic extraction of action items and to-do tasks from email conversations	Email To-Do dataset	ROUGE scores, action-item identification performance	Supports automatic extraction of action items, follow-up tasks, and pending responsibilities from project-related emails	[5]
2023	AI-enabled project management review	Maps AI use across project-management performance areas	Systematic review of published AI and project-management studies	Systematic review screening and thematic analysis	Supports AI use for planning, monitoring, measurement and decision support	[6]

Year	Technology / Study	Application	Dataset	Evaluation Metrics	Relevance to Proposed Project	Ref.
2022	Microsoft Outlook Object Model	Provides programmatic access to Outlook folders, mail items, properties, methods and events	Microsoft Outlook desktop objects and documentation	Not applicable; technical platform documentation	Supports automatic email detection and controlled communication with Outlook 2016	[7]

2.2 RELATED WORK

MailEx shows that events, people, actions, and related details can be extracted from conversational email threads [1]. It also confirms that thread history matters because one project instruction may be spread across several replies. This supports the proposed system’s use of thread-level analysis instead of processing every message as an isolated email.

The Advanced Messaging Platform combines email parsing, intent recognition, entity extraction, action handling, and human validation in an enterprise workflow [2]. It is the closest recent example of a complete AI email pipeline, but it focuses on financial response tasks rather than project meetings, deadlines, Excel archival, and post-action follow-up.

Recent work on temporal normalization, action item extraction, and meeting record generation supports several functions required by this project. Temporal normalization improves interpretation of relative deadlines and meeting dates, while action item extraction techniques support identification of responsibilities, pending tasks, and follow-up activities within project-related emails. MeetingBank further supports structured meeting records and post-meeting documentation, while research on AI-enabled project management justifies the use of AI for planning, monitoring, and decision support [3]-[6].

The research gap is practical. Existing studies solve email extraction, date understanding, action item extraction, meeting record generation, or project-management support separately. They do not provide one stand-alone desktop system that connects Outlook 2016 with Excel, classifies project communication, extracts pending work items, tracks minutes of meeting, reports, presentations, and departmental inputs, and generates timely alerts without changing the manager's working style.

2.3 COMPARISON TABLE WITH EXISTING STUDIES

Table 2.3.1: Comparison with Existing Studies and Systems

Study / System	Email Intent & Context	Meeting / Deadline / Action Extraction	Thread-Aware Follow-Up	Outlook–Excel Archive	Dashboard & Alerts	Main Limitation
MailEx [1]	Yes	Yes	No	No	No	Research dataset and extraction task only
AMP [2]	Yes	Yes	Partial	No	Partial	Built for financial email response workflows
Temporal Normalization [3]	No	Dates and time only	No	No	No	Solves one language-processing task

Study / System	Email Intent & Context	Meeting / Deadline / Action Extraction	Thread-Aware Follow-Up	Outlook–Excel Archive	Dashboard & Alerts	Main Limitation
MeetingBank / Focus on the Action [4], [5]	Meeting context	Meeting records and action-item extraction	No	No	No	Focuses on meeting records and email action-item extraction rather than complete Outlook-integrated project management workflows
Proposed AI-Enabled PM Software	Yes, across email threads	Yes, including actioner, origin, timeline and documents	Yes: revised dates, priority, missing MoM/reports/inputs	Yes	Yes, with dashboard search and local users	Focused on English Outlook 2016 project workflows in the first release

2.4 CHAPTER SUMMARY

The literature supports a hybrid architecture that combines NLP-based classification and extraction with rule-based business validation. Email event extraction provides the basis for identifying actions, temporal research supports deadline interpretation, meeting studies support record summarization and information retrieval, and project-management research supports planning and monitoring.

PROJECT METHODOLOGY

This chapter describes how the AI-enabled project management software will process email, extract project information, maintain the archive, and support managers through the dashboard and alerts.

3.1 PROPOSED METHODOLOGY (FRAMEWORK / ARCHITECTURE)

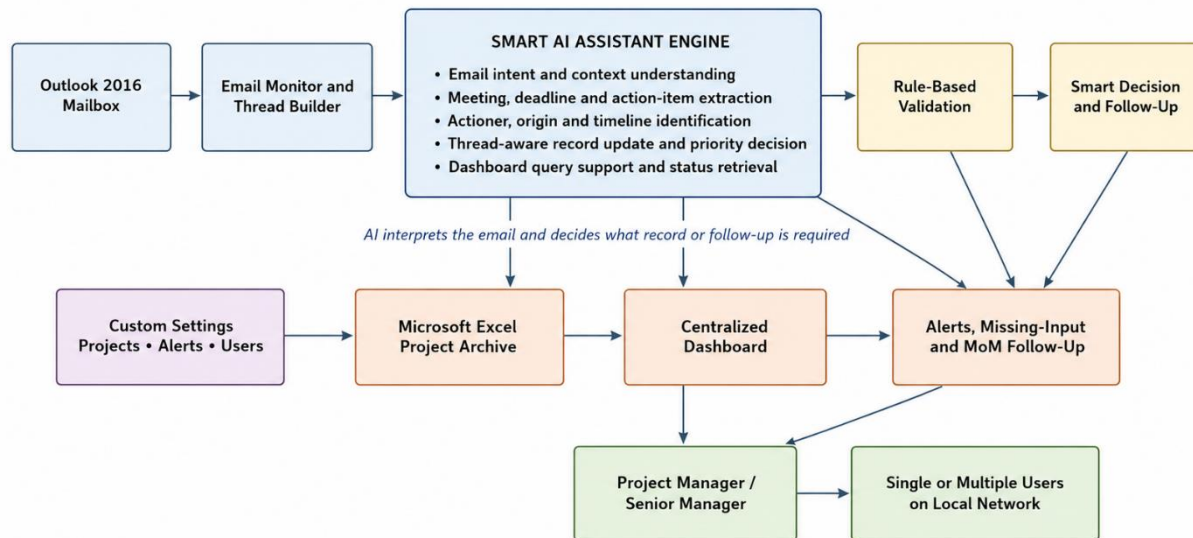
The proposed system uses a hybrid architecture combining NLP-based processing with rule-based validation. Outlook 2016 provides the email subject, sender, recipients, message body, received date, conversation history, and attachment names. The email monitoring module groups related messages into a thread and passes the cleaned content to the AI processing module.

The AI processing module implements the project's NLP functionality by performing email classification, information extraction, named entity recognition, thread analysis, and priority assignment. A lightweight NLP pipeline combining email classification, named entity recognition, and information extraction will process the email's content and context. It will classify whether the message relates to a meeting, deadline, report request, departmental input, follow-up, approval, or general project update. It will then extract the origin, actioner, timeline, meeting details, agenda, requested documents, and action items.

The system will also compare new extractions against the existing archive. If a later email changes a meeting time or deadline, the existing record will be updated. If a meeting has finished but its minutes are missing, the item will be marked for follow-up. If several actions are pending, the system will assign priority and place overdue or near-due items higher on the dashboard. This thread-aware update logic is what distinguishes the system from a simple email parser.

A distinct rule-based validation layer will verify fields that require exact handling, including Outlook metadata, dates, attachment names, known document terms, and user-defined reminder settings. The verified result will be written to Microsoft Excel and presented through the

centralized desktop dashboard. The manager will be able to review records, change a status, adjust settings, and search or filter the stored project archive.



Email content, records and AI processing remain inside the organization's local environment.

Figure 1: System Architecture

3.2 PROCESS MODEL

The project will follow an incremental development model. The system includes Outlook communication, NLP-based email processing, rule-based validation, a dashboard, alerts, and local multi-user access. Building these parts in increments allows the team to produce a working release early, collect feedback, and improve the next release without waiting until the final month.

Increment 1 will finalize the workflow, email types, required fields, interface layout, and initial prototype. Increment 2 will connect Outlook 2016 with the Excel archive. Increment 3 will add the AI processing module and rule validation engine. Increment 4 will add the dashboard, alerts, custom settings, and dashboard search functionality. Increment 5 will add local multi-user access, complete integration, testing, deployment, and documentation.

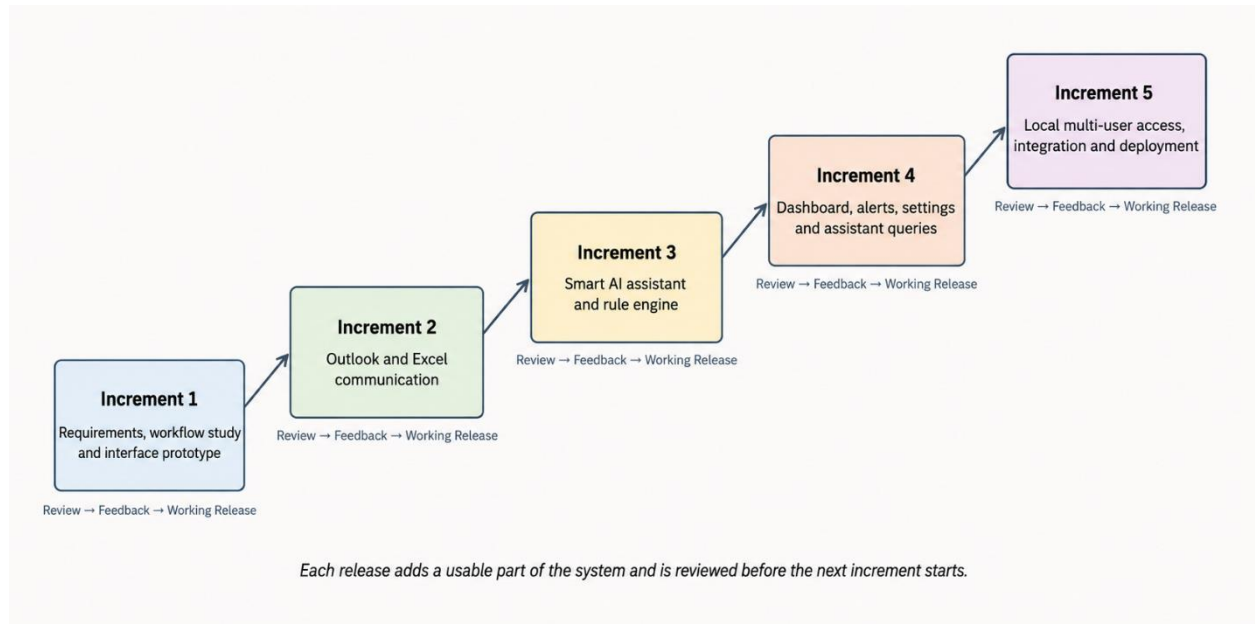


Figure 2: Incremental Development Model for the Proposed System

3.3 MODULES DISCUSSION

Outlook Communication Module: Monitors selected Outlook 2016 folders, reads new messages and thread history, and passes relevant metadata and content to the AI processing module.

AI Processing Module: Classifies email intent, extracts meeting and action information through named entity recognition and information extraction, and supports structured status retrieval from the project archive.

Rule Validation Module: Confirms exact fields, updates revised dates, links actions with responsible people or departments, checks missing post-action documents, and assigns priority for follow-up.

Excel Archive and Data-Management Module: Stores the formal record of meetings, deadlines, actions, documents, alerts, users, and status history in Microsoft Excel.

Centralized Dashboard Module: Shows upcoming meetings, urgent deadlines, overdue items, pending inputs, missing minutes of meeting or reports, recent updates, search, filters, and project status summaries.

Reminder Engine: Creates alerts before deadlines and prompts the manager when required documents or departmental inputs remain pending.

Custom Settings and User Module: Controls projects, monitored folders, reminder times, document types, user roles, and single-user or local-network operation.

Security and Backup Module: Keeps access under user-defined permissions and maintains local backups of the archive and configuration.

3.4 CHAPTER SUMMARY

The proposed design clearly separates the NLP-based processing from traditional software functions. Outlook and Excel handle communication and archival. The AI processing module classifies emails, extracts project information, detects changes and pending actions, and supports information retrieval. The rule validation layer, dashboard, reminder engine, settings, and local-network support turn this processing capability into a usable project-management tool.

PROJECT SCHEDULE

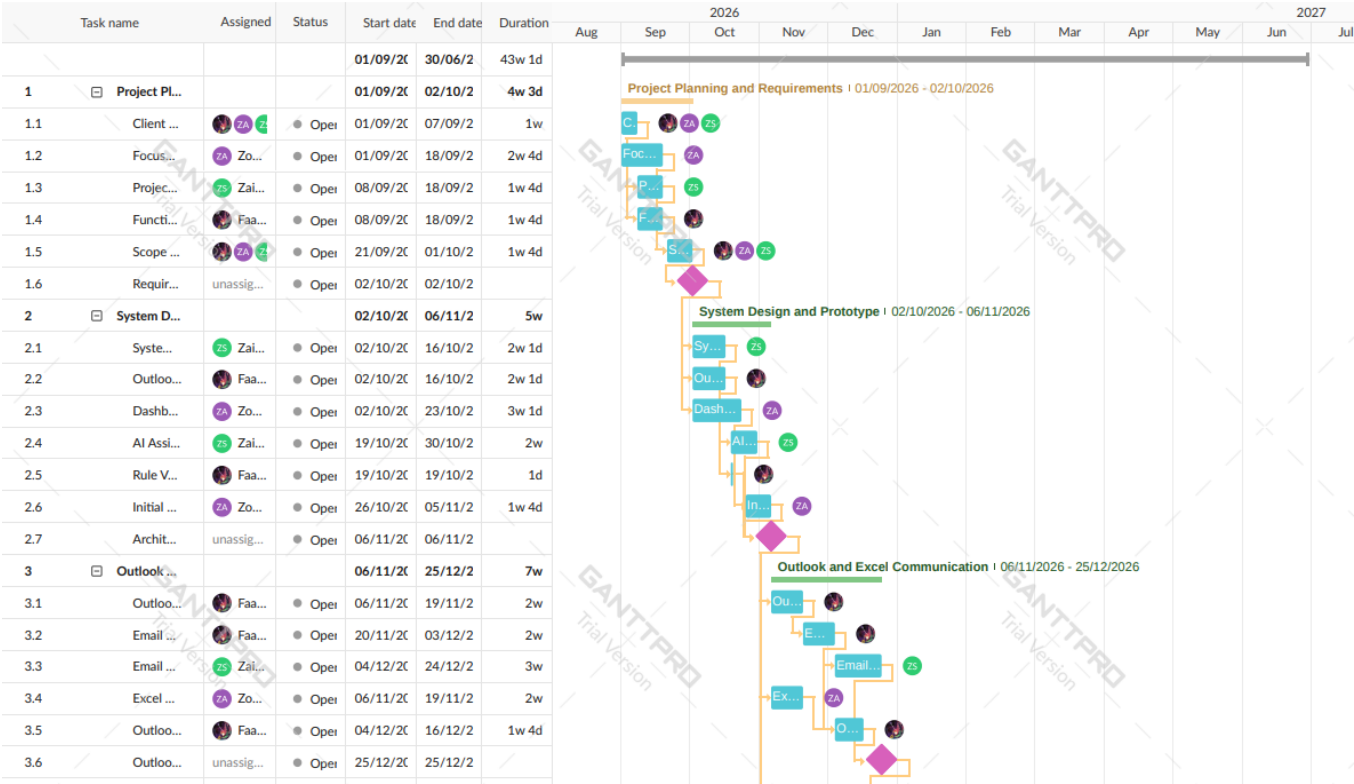
KEY MILESTONES

The schedule covers ten months, or approximately forty working weeks. The milestone table shows the main outputs expected during each stage.

S. No	Milestone	Expected Completion
M1	Scope refinement and proposal revision	Month 1 (Week 2)
M2	Focused literature review	Month 1 (Week 4)
M3	Project use cases, email types and required data fields	Month 2 (Week 6)
M4	System architecture, interface wireframes and incremental plan	Month 2 (Week 8)
M5	Outlook 2016 email detection and thread-reading prototype	Month 3 (Week 11)
M6	Microsoft Excel archive and formal project log	Month 4 (Week 13)
M7	Email classification and information extraction prototype	Month 4 (Week 15)
M8	Meeting, deadline, actioner and action-item extraction	Month 5 (Week 19)
M9	Thread-aware updates, priority assignment and follow-up logic	Month 6 (Week 22)
M10	Centralized dashboard prototype	Month 6 (Week 21)
M11	Reminder and alert engine	Month 6 (Week 24)
M12	Dashboard search, filtering and project status summaries	Month 7 (Week 27)
M13	Custom settings and user preferences	Month 7 (Week 28)
M14	Local network and multi-user access	Month 8 (Week 31)
M15	Security, access control and backup handling	Month 8 (Week 32)
M16	Complete system integration	Month 9 (Week 35)
M17	Pilot testing, feedback and final refinement	Month 10 (Week 38)
M18	Final documentation, deployment package and viva preparation	Month 10 (Week 40)

GANTT CHART

My Team | Outlook Integrated AI Project Management Dashboard



4	Smart AI ...		06/11/2026	15/02/27	14w 1d
4.1	Select ...	Oper	06/11/2026	19/11/26	2w
4.2	Prepar...	Oper	20/11/2026	17/12/26	4w
4.3	Email L...	Oper	20/11/2026	17/12/26	4w
4.4	Meeti...	Oper	18/12/2026	07/01/27	3w
4.5	Action...	Oper	08/01/2027	21/01/27	2w
4.6	Rule V...	Oper	25/12/2026	14/01/27	3w
4.7	Attach...	Oper	15/01/2027	12/02/27	4w 1d
4.8	Thread...	Oper	08/01/2027	12/02/27	5w 1d
4.9	Smart ...	Oper	15/02/2027	15/02/27	
5	Dashboar...		25/12/2026	15/03/27	11w 1d
5.1	Dashb...	Oper	25/12/2026	14/01/27	3w
5.2	Meeti...	Oper	15/01/2027	04/02/27	3w
5.3	Search...	Oper	05/02/2027	12/03/27	5w 1d
5.4	Remin...	Oper	15/01/2027	03/02/27	2w 4d
5.5	MoM...	Oper	04/02/2027	17/02/27	2w
5.6	Custo...	Oper	05/02/2027	17/02/27	1w 4d
5.7	Dashb...	Oper	15/03/2027	15/03/27	
6	Multi-Us...		15/03/2027	23/04/27	5w 4d
6.1	Local ...	Oper	15/03/2027	26/03/27	2w
6.2	User R...	Oper	29/03/2027	09/04/27	2w
6.3	Backu...	Oper	15/03/2027	26/03/27	2w
6.4	Integra...	Oper	15/03/2027	09/04/27	4w
6.5	Multi...	Oper	12/04/2027	22/04/27	1w 4d
6.6	Compl...	Oper	23/04/2027	23/04/27	

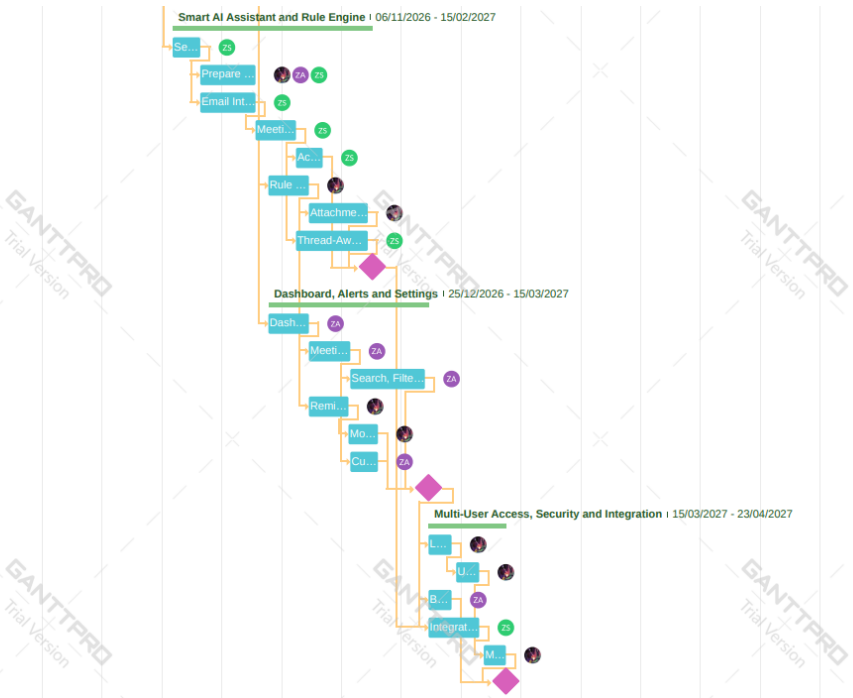




Figure 3: Proposed 10-Month / 40-Week Gantt Chart with Team Responsibilities

RESOURCE ESTIMATION

The project is software-based. Most development tools are already available or open source, so the estimated cost mainly covers backup storage, local-network testing, printing, and final documentation.

Item	Specification / Purpose	Estimated Cost (PKR)
Development laptops	Existing Windows laptops for software development, NLP module testing and documentation	0 (already owned)
Microsoft Office 2016	Existing Outlook 2016 and Excel installation for email communication and archival	0 (available)
Python and development libraries	Desktop interface, Outlook communication, Excel handling, AI/NLP, alerts and local-network modules	0 (open source)
Local network equipment	Existing router and local network for multi-user testing	0 (available)
Backup storage	External drive or secure local backup for project files and Excel archive	6,000–8,000
Printing and binding	Proposal, progress material, final report and presentation documents	4,000–6,000
Contingency	Minor testing and documentation expenses	2,000–3,000
Total Estimated Cost		12,000–17,000

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